Modeling Neutron Stars
Lecture-3

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UNIVERSITÉ CAEN NORMANDIE

Nuclear Theory Group
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Contents:

Elementary Field Theory

- Recap
- Lagrangian density and functionals
- Classical Field Theory
- Transformations

Action Principle

Particle mechanics

- An action and Lagrangian is defined for an event happening in between time t_1 and t_2 as,

$$\mathscr{A} = \int_{t_1}^{t_2} dt\, \mathrm{L}(q_r(t), \dot{q}_r(t), t).$$

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$$\boxed{\frac{d}{dt}\left(\frac{\partial \mathrm{L}}{\partial \dot{q}_r}\right) = \frac{\partial \mathrm{L}}{\partial q_r}}$$

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$\mathscr{L}$ and $\mathscr{A}$ are Poincaré invariant *i.e.* invariant under translation and Lorentz transformation.

Functional and its derivative

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\text{So, } \frac{\delta F\,[\,f\,]}{\delta f(x_0)} \quad &= \quad \lim_{\epsilon \to 0} \frac{F\,[\,f + \epsilon\delta(x - x_0)] - F\,[\,f\,]}{\epsilon}
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- **Problem 1(a): Using the definition of functional derivative find the derivative for $F[f] = \int_{-1}^{1} f(x)\ dx$.**

  (b): Find $\frac{\delta H}{\delta f(z)}$ for $H[\ f\ ] = \int_a^b G(x,y) f(y) dy$.

  (c): Find $\frac{\delta^2 F}{\delta f(x_0)\delta f(x_1)}$ for $F[f^{\,3}]$ for the same f as part (a).

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- $\mathscr{L}$ is a function of Φ_{A} and $\partial_\mu \Phi_{\mathrm{A}}$, however, $\mathscr{A}$ or L are *functionals* for different Φ_{A} and $\partial_\mu \Phi_{\mathrm{A}}$ at a given t.

Euler-Lagrange equations for fields

Principle of least action

- Variation on the field Φ^{A} and its covariant derivative $\partial_\mu \Phi^{\mathrm{A}}$ are given by

$$\begin{aligned} \Phi^{\mathrm{A}}(x) &\rightarrow \Phi^{\mathrm{A}}(x) + \delta\Phi^{\mathrm{A}}(x), \\ \partial_\mu \Phi^{\mathrm{A}}(x) &\rightarrow \partial_\mu \Phi^{\mathrm{A}}(x) + \partial_\mu \delta\Phi^{\mathrm{A}}(x). \end{aligned}$$

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In the last line we invoked the Gauss theorem in *four dimension*. $\delta\Phi^{\mathrm{A}}$ vanishes everywhere on the boundary $\partial\Omega$.

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- **Problem 2: Find the Euler Lagrange equations for the following cases**
  (a) For a scalar field ϕ where the Lagrangian is given by $\mathscr{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2} m^2 \phi^2 - V(\phi)$.

  (b) For a complex scalar field ϕ with the Lagrangian $\mathscr{L} = (\partial_\mu \phi)^\dagger (\partial^\mu \phi) - m^2 \phi^\dagger \phi - V(\phi^\dagger \phi)$.

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  Note: ϕ and $\phi^\dagger$ are independent!!

Euler-Lagrange eqs for electromagnetic field

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&= 2(\partial^\mu A^\nu \partial_\mu A_\nu - \partial^\mu A^\nu \partial_\nu A_\mu)
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- Remember the inhomogeneous Maxwell's equations from last class!!

Conjugate momentum

- Same as the case of particle mechanics, momentum canonical to the field Φ^{A} is defined as

$$\Pi_{\mathrm{A}} \equiv \frac{\delta L}{\delta \dot{\Phi}^{\mathrm{A}}}$$

Remember, L is a *functional*!! So the derivative is a functional derivative. The building blocks of L are Φ^{A}, $\dot{\Phi}^{\mathrm{A}}$ and $\boldsymbol{\nabla}\Phi^{\mathrm{A}}$.

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$$\frac{\delta}{\delta \dot{\Phi}^{\mathrm{A}}(t, \boldsymbol{x})} \dot{\Phi}^{\mathrm{B}}(t, \boldsymbol{y}) = \delta^{\mathrm{A}}_{\mathrm{B}} \delta^3(\boldsymbol{x} - \boldsymbol{y}).$$

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  Notice this is not dimensionless, rather has dimension of inverse volume.

Hamiltonian formalism

- Using the analogy of particle mechanics, Hamiltonian (density!!! remember!!!) can be written as

$$\mathscr{H}(\Phi^{\mathrm{A}}, \boldsymbol{\nabla}\Phi^{\mathrm{A}}, \Pi_{\mathrm{A}}) = \Pi_{\mathrm{A}}\dot{\Phi}^{\mathrm{A}} - \mathscr{L}(\Phi^{\mathrm{A}}, \partial_\mu \Phi^{\mathrm{A}}).$$

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- **Problem 3: Show that, $[\Phi^{\mathrm{A}}(\boldsymbol{t},\boldsymbol{x}), \Pi_{\mathrm{B}}(\boldsymbol{t},\boldsymbol{x})]_{\mathrm{P}} = \delta^{\mathrm{A}}_{\mathrm{B}} \delta^3(\boldsymbol{x}-\boldsymbol{y})$.**

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- **Problem 4: Do the same exercise for the cases of complex scalar field and the electromagnetic field.**

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- **Noether's theorem** does this job. The conserved quantities are referred as *Noether charge* or *Noether current*.

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- Here $\Omega \rightarrow \Omega'$ for the co-ordinate transformation $x \rightarrow x'$.

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$$\begin{aligned} x^\mu &\rightarrow x'^\mu = x^\mu + \delta x^\mu. \\ \Rightarrow \Phi^{\mathrm{A}}(x) &\rightarrow \Phi'^{\mathrm{A}}(x') = \Phi^{\mathrm{A}}(x) + \delta\Phi^{\mathrm{A}}(x). \end{aligned}$$

- The change in the action because of this transformation is given by,

$$\delta\mathscr{A} = \int_{\Omega'} d^4x' \mathscr{L}\left(\Phi'^{\mathrm{A}}(x'), \partial'_\mu \Phi'^{\mathrm{A}}(x')\right) - \int_{\Omega} d^4x \mathscr{L}\left(\Phi^{\mathrm{A}}(x), \partial_\mu \Phi^{\mathrm{A}}(x)\right).$$

- Here $\Omega \rightarrow \Omega'$ for the co-ordinate transformation $x \rightarrow x'$.

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Noether's Theorem

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Noether's Theorem

Continued.....

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$$\begin{aligned}\overline{\delta}f(x) &\equiv f'(x) - f(x) \\ &= \left[f'(x') - f(x)\right] - \left[f'(x') - f'(x)\right] \\ &= \delta f(x) - \partial_\mu f(x)\delta x^\mu.\end{aligned}$$

- Notice $\overline{\delta}$ commutes with ∂_μ.

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Continued.....

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&= \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\right) \overline{\delta}\Phi^{\mathrm{A}} + \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial_\mu \left(\overline{\delta}\Phi^{\mathrm{A}}\right) \\
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&= \int_\Omega d^4x \partial_\mu \left[\frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \left(\delta \Phi^{\mathrm{A}} - \partial^\nu \Phi^{\mathrm{A}} \delta x_\nu\right) + g^{\mu\nu} \mathscr{L} \delta x_\nu\right]
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Stress-Energy Tensor

From Noether's Theorem

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- This holds for an arbitrary variation of the fields and the co-ordinates, provided equations of motion are satisfied. However, it doesn't ensure $\delta\mathscr{A} = 0$!! It's because the current may not vanish at the boundary.

Noether Current and charge

Conservation law!

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$$\begin{aligned} \frac{dQ}{dt} &= \int d^3x\, \partial_0 J^0 = -\int d^3x\, \boldsymbol{\nabla}\cdot\boldsymbol{J} = 0 \\ &\Rightarrow \quad Q \to \text{conserved.} \end{aligned}$$

References:

- A first book of Quantum Field Theory, CRC Press, **Amitabha Lahiri and Palash Baran Pal**.
- Quantum Field Theory, Cambridge University Press, **Lewis H. Ryder**.